

## **Background**

John the Ripper (JTR) is a password-auditing tool used to test the strength of password hashes and protected files. Johnny is its graphical interface, making JTR easier to use through point-and-click options. In this lab, JTR and Johnny are used to recover the password of an authorized protected PDF and demonstrate the risks of weak passwords.

NetworkWalks provides online tools such as the Hash Calculator and Password Cracker. The Hash Calculator generates a hash from an authorized protected PDF, while the Password Cracker performs a dictionary-based recovery attempt. These tools help learners understand password attacks and the importance of using strong, unique passwords.

## **Lab purpose**

The purpose of both JTR/Johnny and NetworkWalks tools in this exercise is to demonstrate password-security risks in a controlled and authorized lab environment. JTR and Johnny provide a more practical, professional tool-based experience, while the NetworkWalks tools provide a friendly, online alternative.

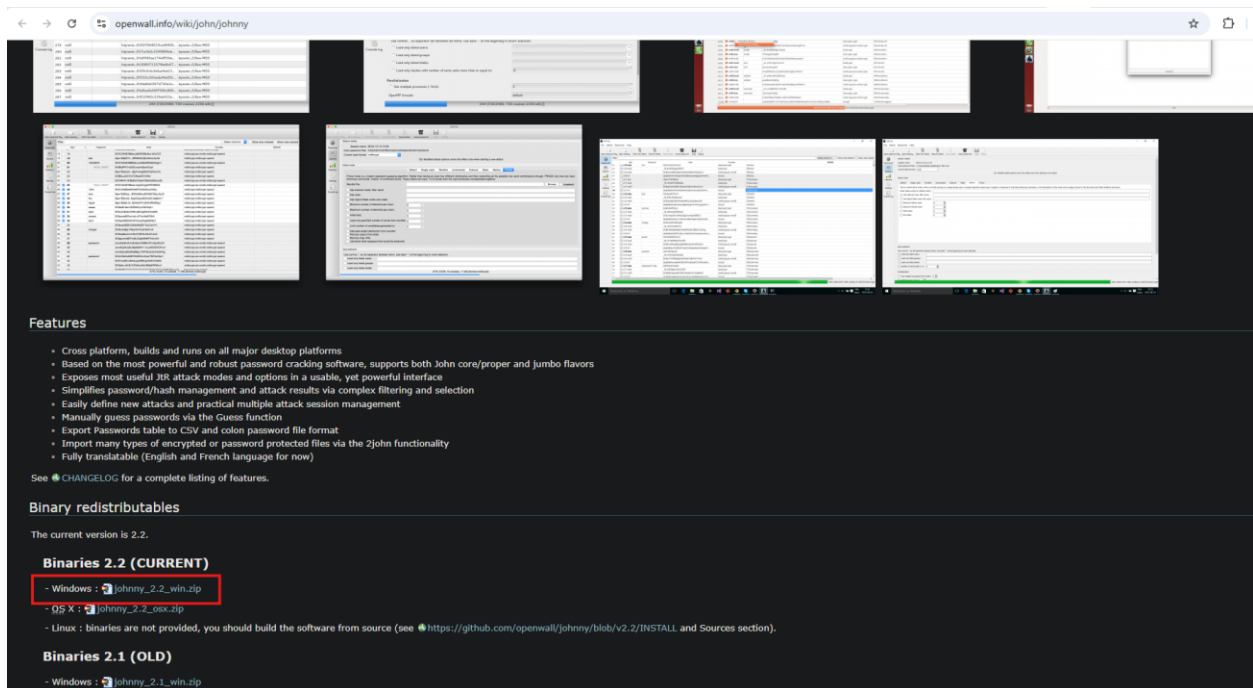
By completing the lab, learners should be able to:

- Understand the concept of password hashes and password-protected PDF files.
- Recognize how dictionary-based password attacks work.
- Use JTR/Johnny or NetworkWalks tools only on authorized lab files.
- Explain why weak, short, reused, or predictable passwords are insecure.
- Recommend strong, unique passphrases and password managers as protective controls

## Task 1

Crack the password of attached PDF file (My Locked PDF3.pdf) using JTR JOHN and JTR JOHNNY tools Windows PC.

Go to <https://openwall.info/wiki/john/johnny> and download [johnny\\_2.2\\_win.zip](#)



The screenshot shows the Openwall website page for John/Johnny. The page features a dark background with a grid of application window screenshots at the top. Below this, the 'Features' section lists various capabilities of the tool. The 'Binary redistributables' section states the current version is 2.2 and lists download links for Windows, OS X, and Linux. The Windows link, 'johnny\_2.2\_win.zip', is highlighted with a red rectangular box. Below this, the 'Binaries 2.1 (OLD)' section lists a Windows download link.

**Features**

- Cross platform, builds and runs on all major desktop platforms
- Based on the most powerful and robust password cracking software, supports both John core/proper and Jumbo flavors
- Exposes most useful JTR attack modes and options in a usable, yet powerful interface
- Simplifies password/hash management and attack results via complex filtering and selection
- Easily define new attacks and practical multiple attack session management
- Manually guess passwords via the Guess function
- Export Passwords table to CSV and colon password file format
- Import many types of encrypted or password protected files via the 2john functionality
- Fully translatable (English and French language for now)

See CHANGELOG for a complete listing of features.

**Binary redistributables**

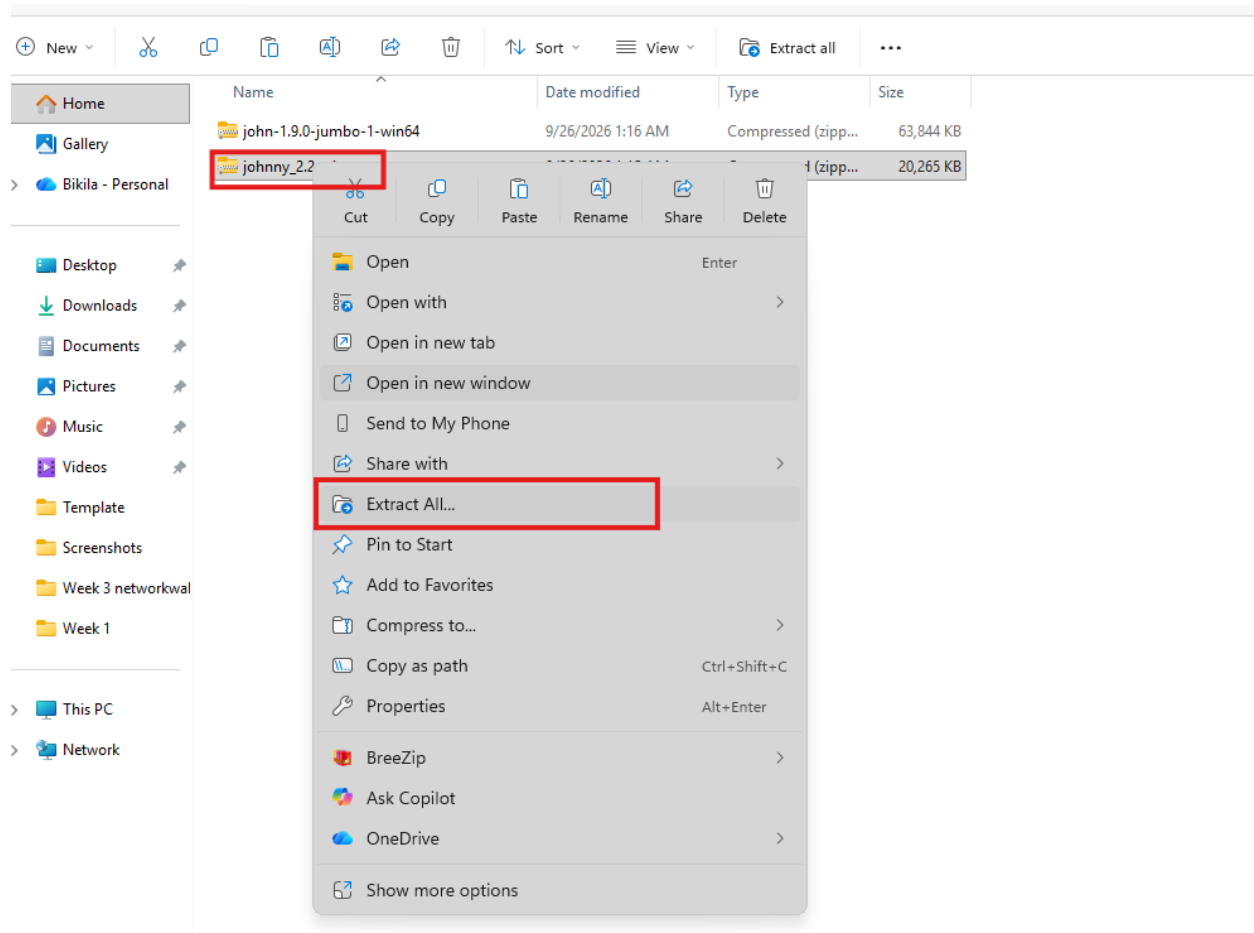
The current version is 2.2.

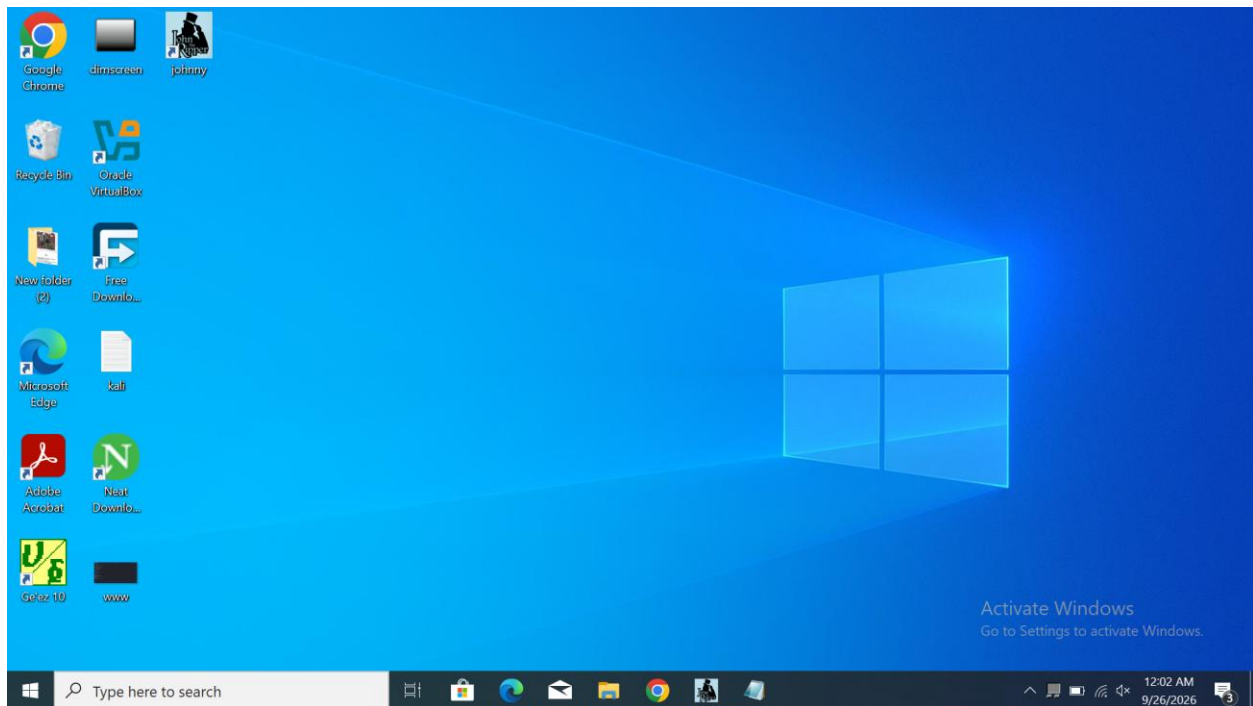
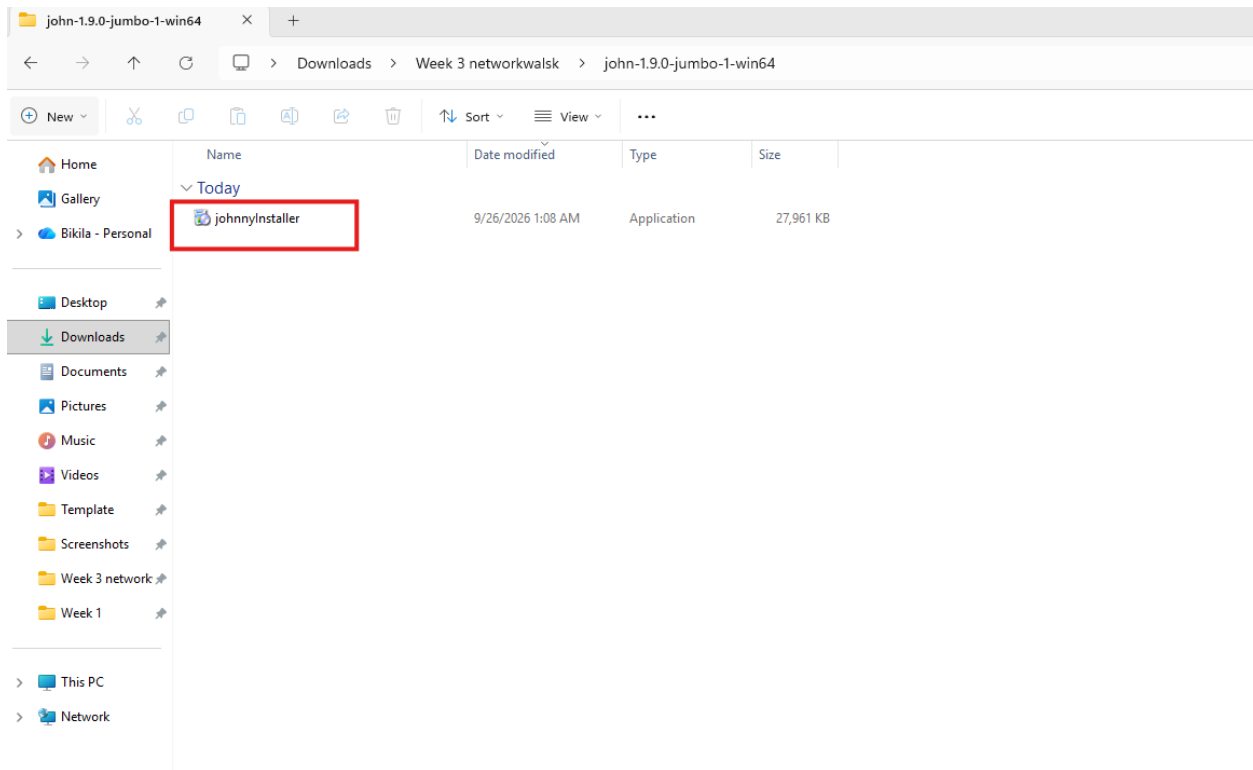
**Binaries 2.2 (CURRENT)**

- Windows : [johnny\\_2.2\\_win.zip](#)
- OS X : [johnny\\_2.2\\_osx.zip](#)
- Linux : binaries are not provided, you should build the software from source (see <https://github.com/openwall/johnny/blob/v2.2/INSTALL> and Sources section).

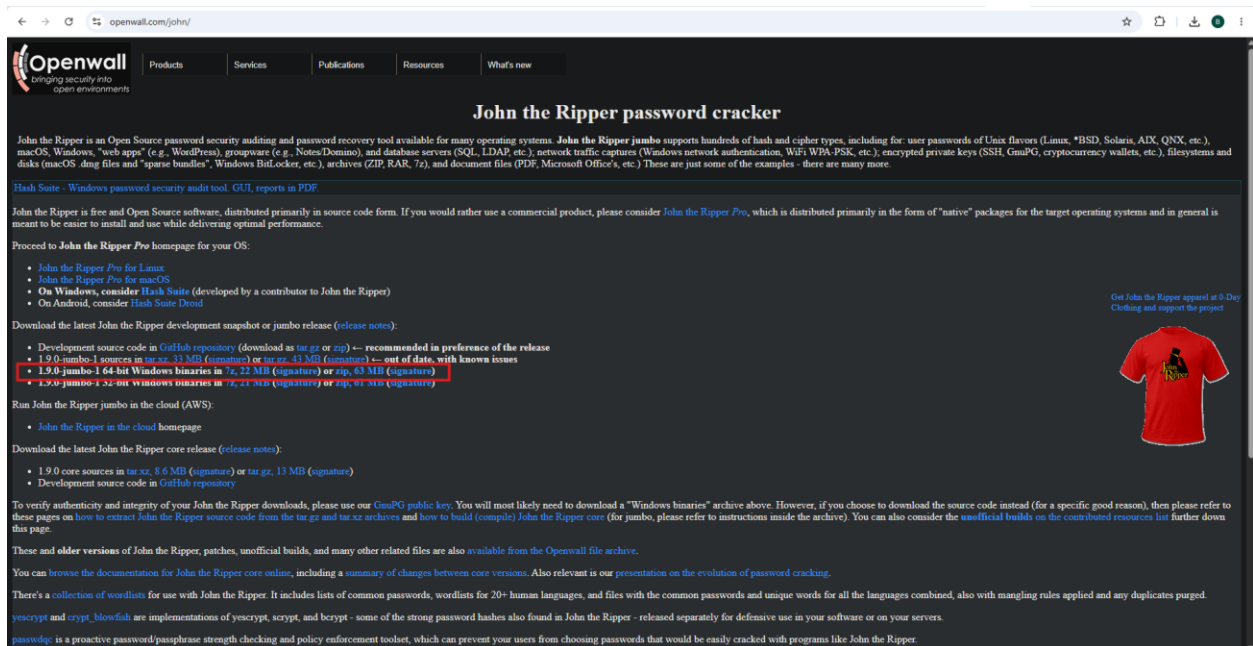
**Binaries 2.1 (OLD)**

- Windows : [johnny\\_2.1\\_win.zip](#)





Go to <https://www.openwall.com/john/> and download [john-1.9.0-jumbo-1-win64.zip](#)



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## John the Ripper password cracker

John the Ripper is an Open Source password security auditing and password recovery tool available for many operating systems. John the Ripper jumbo supports hundreds of hash and cipher types, including for: user passwords of Unix flavors (Linux, \*BSD, Solaris, AIX, QNX, etc.), macOS, Windows, "web apps" (e.g., WordPress), groupware (e.g., Notes/Domino), and database servers (SQL, LDAP, etc.); network traffic captures (Windows network authentication, WiFi WPA-PSK, etc.); encrypted private keys (SSH, GnuPG, cryptocurrency wallets, etc.); filesystems and disks (macOS .dmg files and "sparse bundles", Windows BitLocker, etc.); archives (ZIP, RAR, 7z), and document files (PDF, Microsoft Office's, etc.) These are just some of the examples - there are many more.

[Hash Suite](#) - Windows password security audit tool, GUI, reports in PDF.

John the Ripper is free and Open Source software, distributed primarily in source code form. If you would rather use a commercial product, please consider *John the Ripper Pro*, which is distributed primarily in the form of "native" packages for the target operating systems and in general is meant to be easier to install and use while delivering optimal performance.

Proceed to [John the Ripper Pro](#) homepage for your OS:

- [John the Ripper Pro for Linux](#)
- [John the Ripper Pro for macOS](#)
- [On Windows, consider Hash Suite](#) (developed by a contributor to John the Ripper)
- [On Android, consider Hash Suite Droid](#)

Download the latest John the Ripper development snapshot or jumbo release ([release notes](#)):

- Development source code in [GitHub repository](#) (download as [tar.gz](#) or [zip](#)) ← **recommended in preference of the release**
- [1.9.0-jumbo-1 sources in tar.gz, 13 MB \(signature\) or tar.gz, 13 MB \(checksum\) ← not of date, with known issues](#)
- **[1.9.0-jumbo-1 64-bit Windows binaries in 7z, 22 MB \(signature\) or zip, 63 MB \(signature\)](#)**
- [1.9.0-jumbo-1 32-bit Windows binaries in 7z, 21 MB \(signature\) or zip, 61 MB \(signature\)](#)

Run John the Ripper jumbo in the cloud (AWS):

- [John the Ripper in the cloud homepage](#)

Download the latest John the Ripper core release ([release notes](#)):

- [1.9.0 core sources in tar.gz, 8.6 MB \(signature\) or tar.gz, 13 MB \(signature\)](#)
- Development source code in [GitHub repository](#)

To verify authenticity and integrity of your John the Ripper downloads, please use our [GnuPG public key](#). You will most likely need to download a "Windows binaries" archive above. However, if you choose to download the source code instead (for a specific good reason), then please refer to these pages on [how to extract John the Ripper source code from the tar.gz and tar.gz archives](#) and [how to build \(compile\) John the Ripper core](#) (for jumbo, please refer to instructions inside the archive). You can also consider the [unofficial builds on the contributed resources list](#) further down this page.

These and older versions of John the Ripper, patches, unofficial builds, and many other related files are also available [from the Openwall file archive](#).

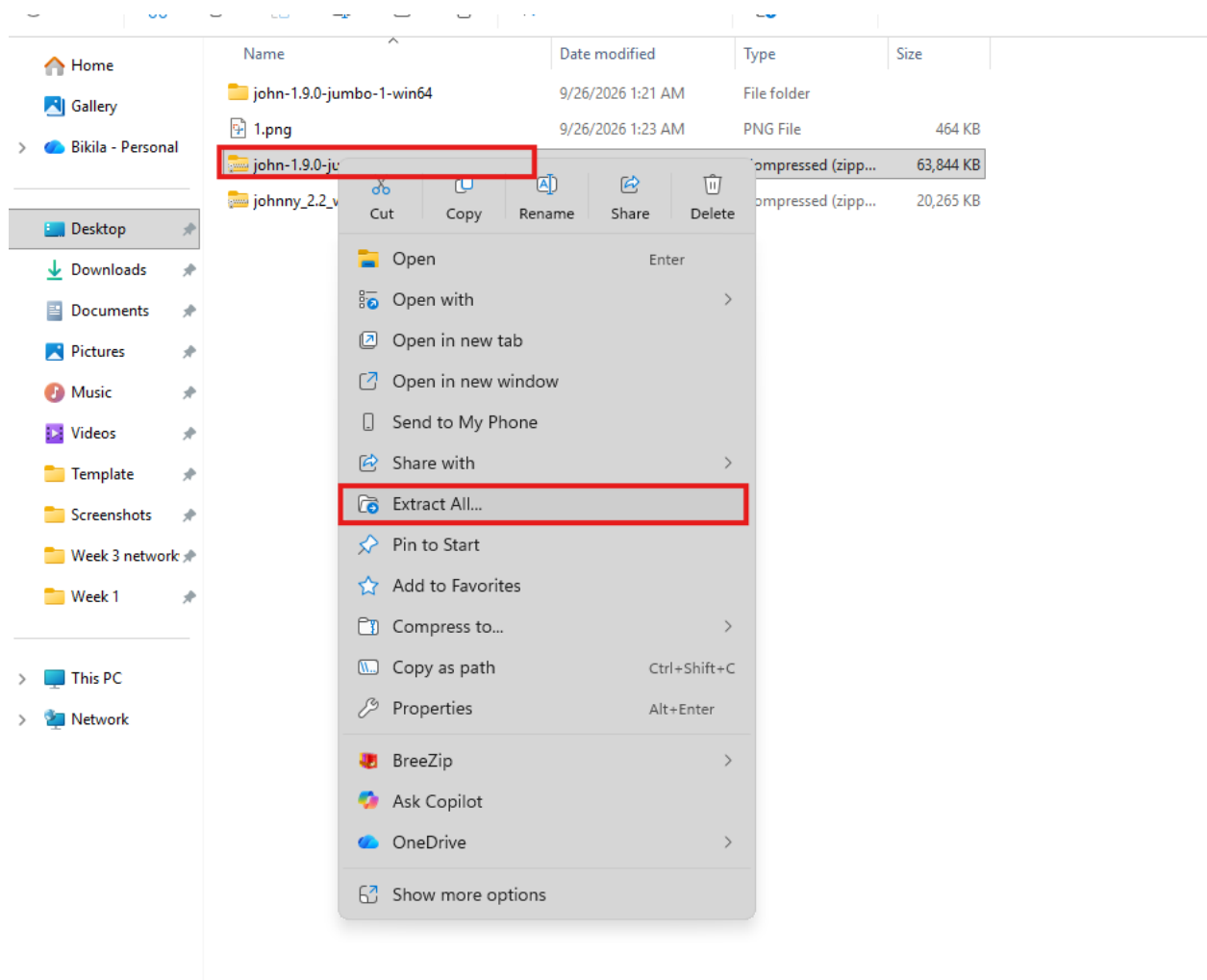
You can [browse the documentation](#) for John the Ripper core online, including a [summary of changes between core versions](#). Also relevant is our [presentation on the evolution of password cracking](#).

There's a [collection of wordlists](#) for use with John the Ripper. It includes lists of common passwords, wordlists for 20+ human languages, and files with the common passwords and unique words for all the languages combined, also with mangling rules applied and any duplicates purged.

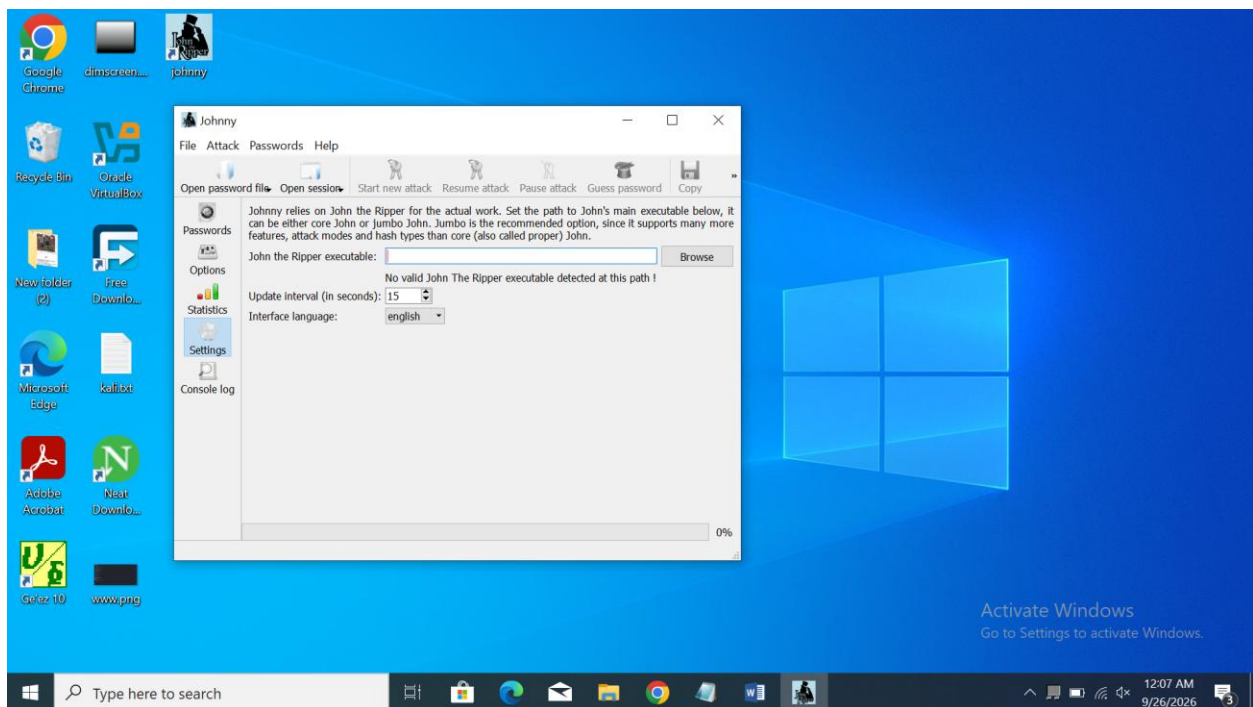
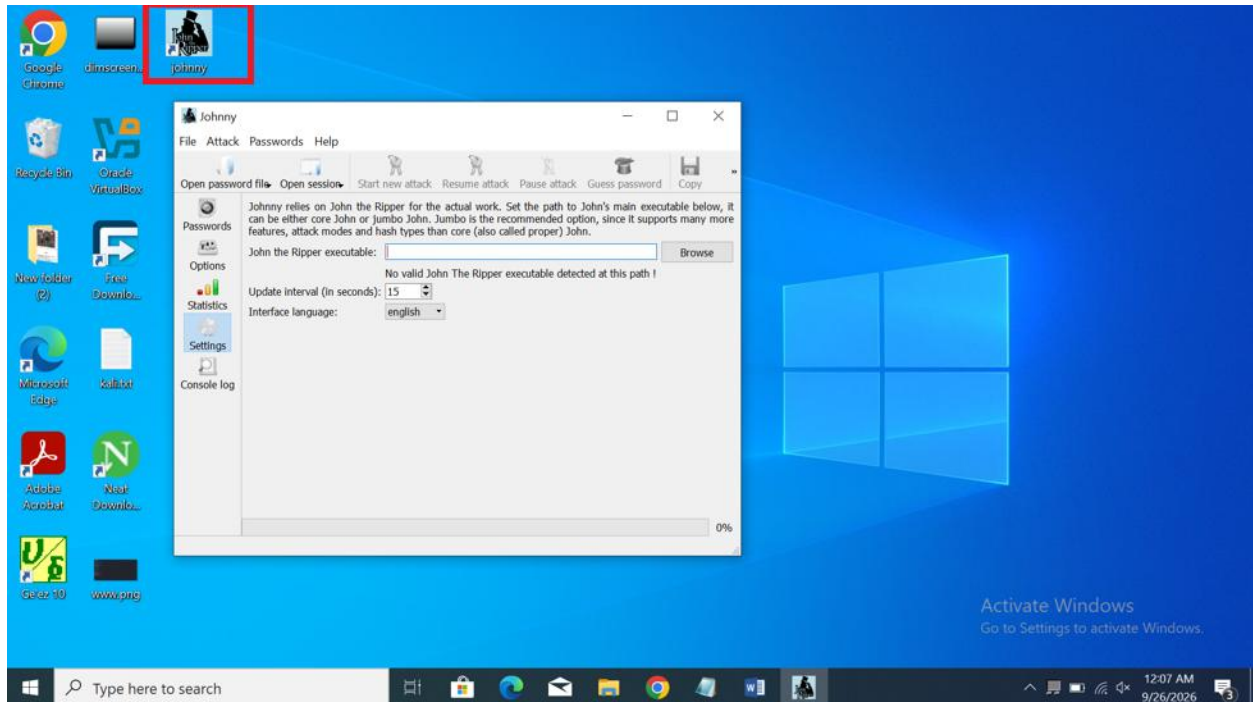
[yescrypt](#) and [crypt\\_mowfish](#) are implementations of yescrypt, script, and bcrypt - some of the strong password hashes also found in John the Ripper - released separately for defensive use in your software or on your servers.

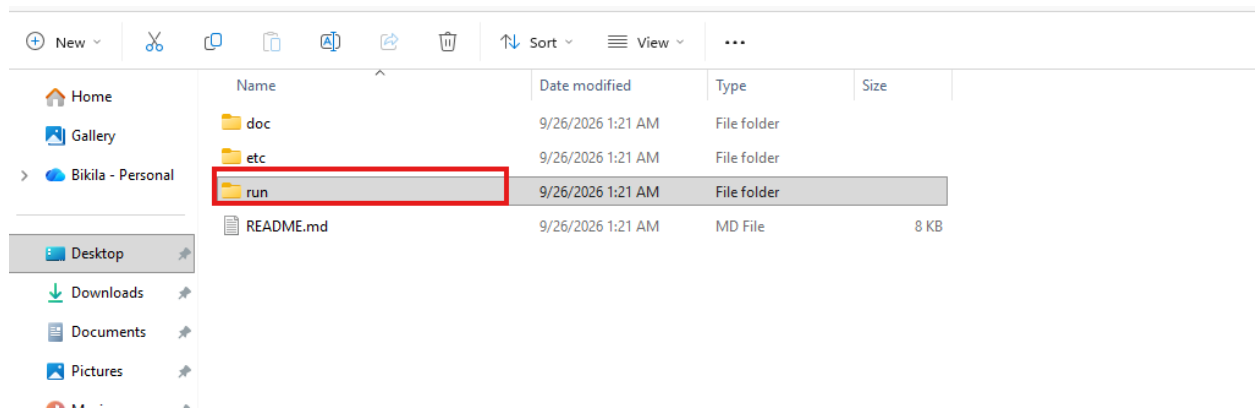
[passwords](#) is a proactive password/passphrase strength checking and policy enforcement toolset, which can prevent your users from choosing passwords that would be easily cracked with programs like John the Ripper.

## Extract john-1.9.0-jumbo-1-win64.zip

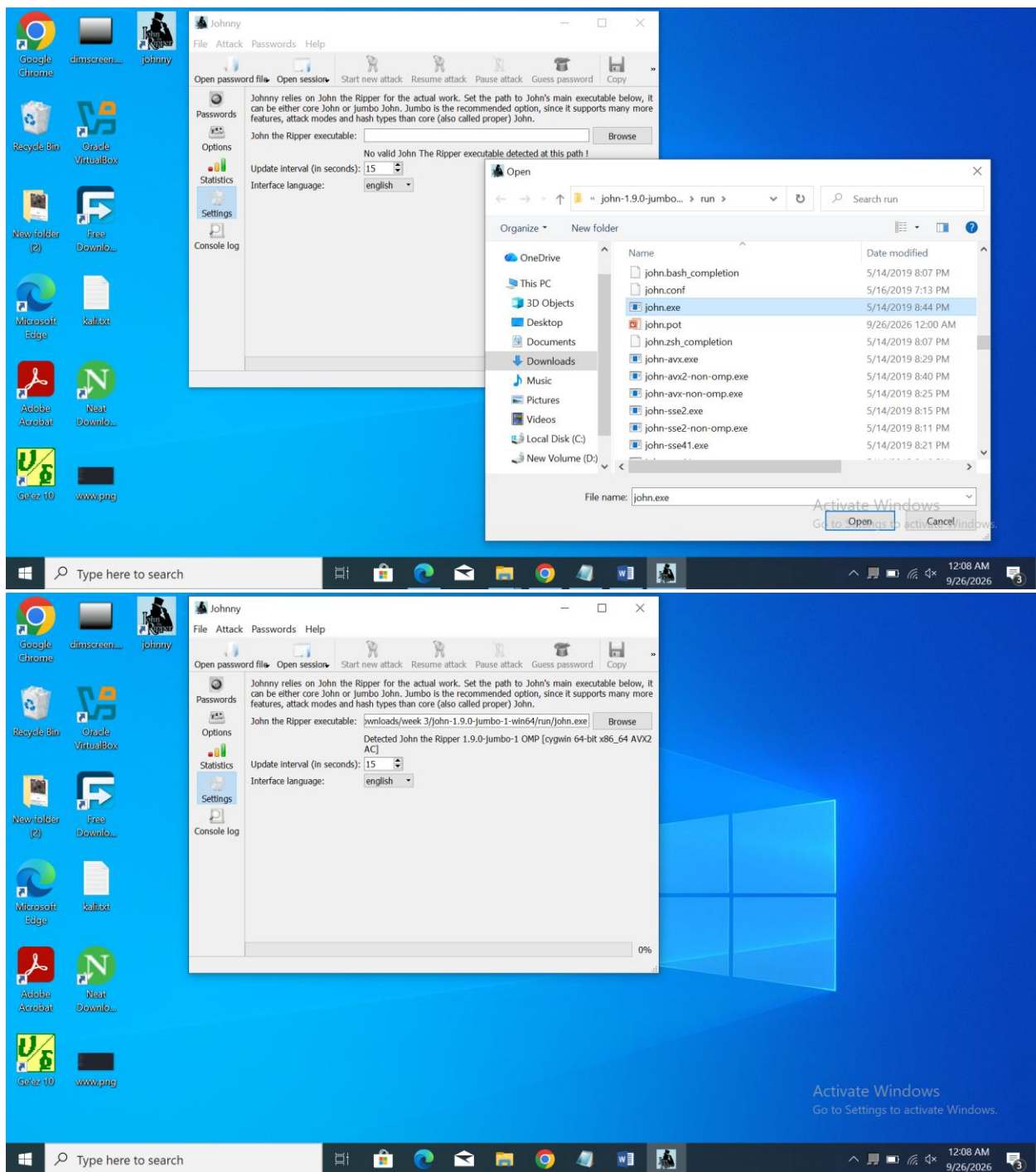


## Open johnny application



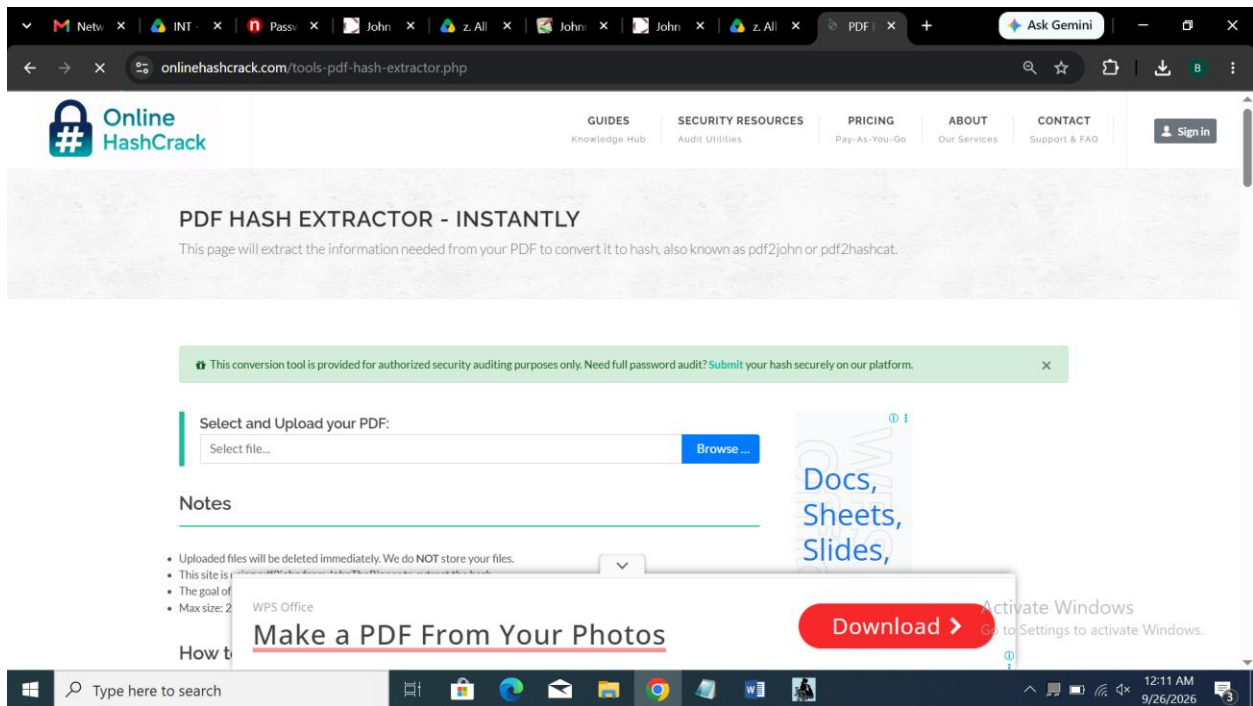


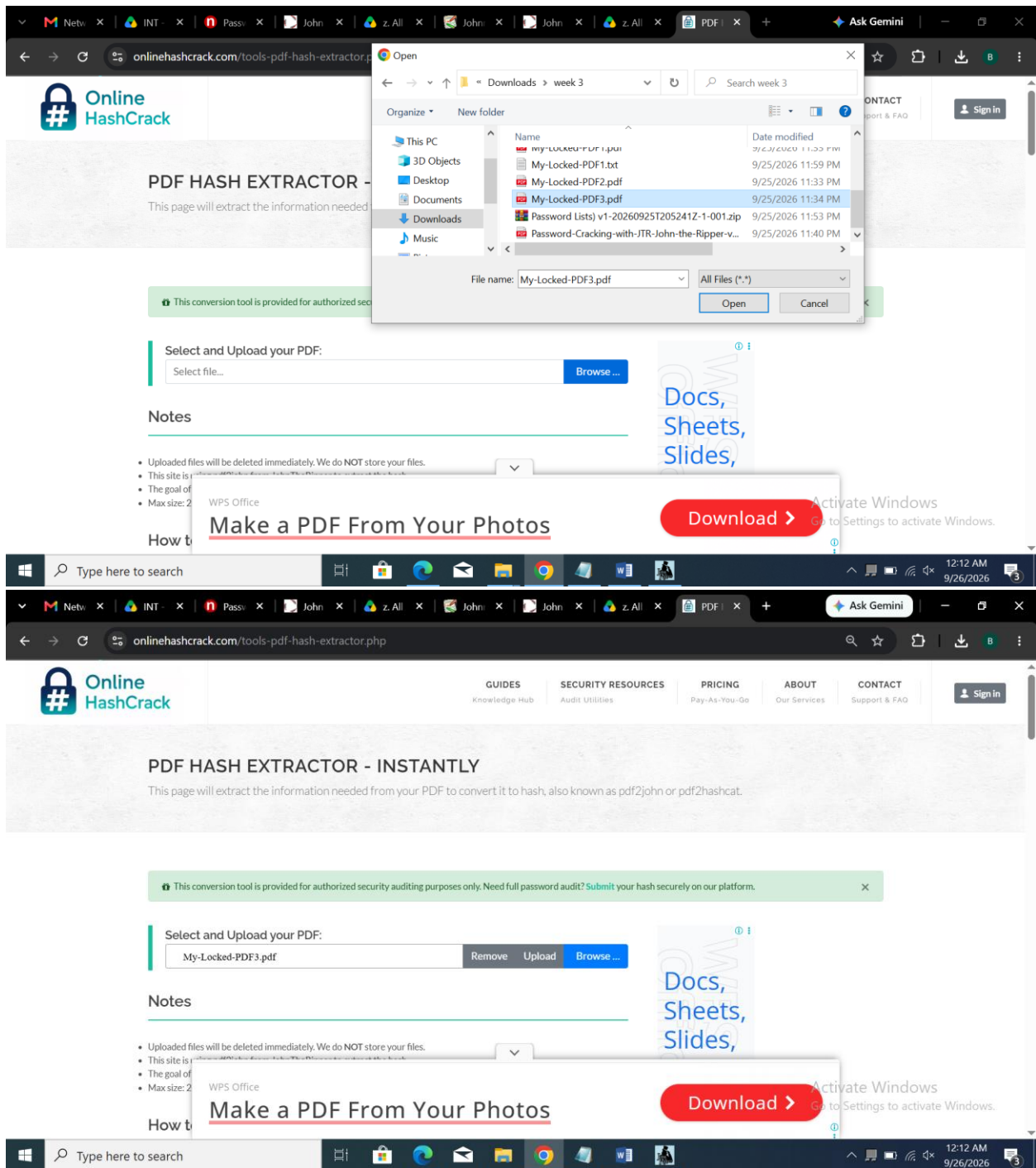




Open the hash website & upload your pdf file to find its hash:

<https://www.onlinehashcrack.com/tools-pdf-hash-extractor.php>





Netw x INT x Passv x John x z.All x John x John x z.All x PDF x + Ask Gemini

onlinehashcrack.com/tools-pdf-hash-extractor.php

Online HashCrack

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### PDF HASH EXTRACTOR - INSTANTLY

This page will extract the information needed from your PDF to convert it to hash, also known as pdf2john or pdf2hashcat.

This conversion tool is provided for authorized security auditing purposes only. Need full password audit? Submit your hash securely on our platform.

Select and Upload your PDF:

Select file... Remove Upload Browse...

Output:

```
$pdf$4*4*128*1028*1*16*34eb542ef4e1b0b32d25ce15a9a7281*32*b77872bfc9a24fb284506628  
3a8fc1b0021446990b9e4114071a4d9104984c1*32*e7572256e4b552cd57988f5134214b91920d94  
d7a6bf550ea94a2995c7f2ab02
```

Notes

Docs, Sheets, Slides, PDF: One

Activate Windows  
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- Readme - in Crack
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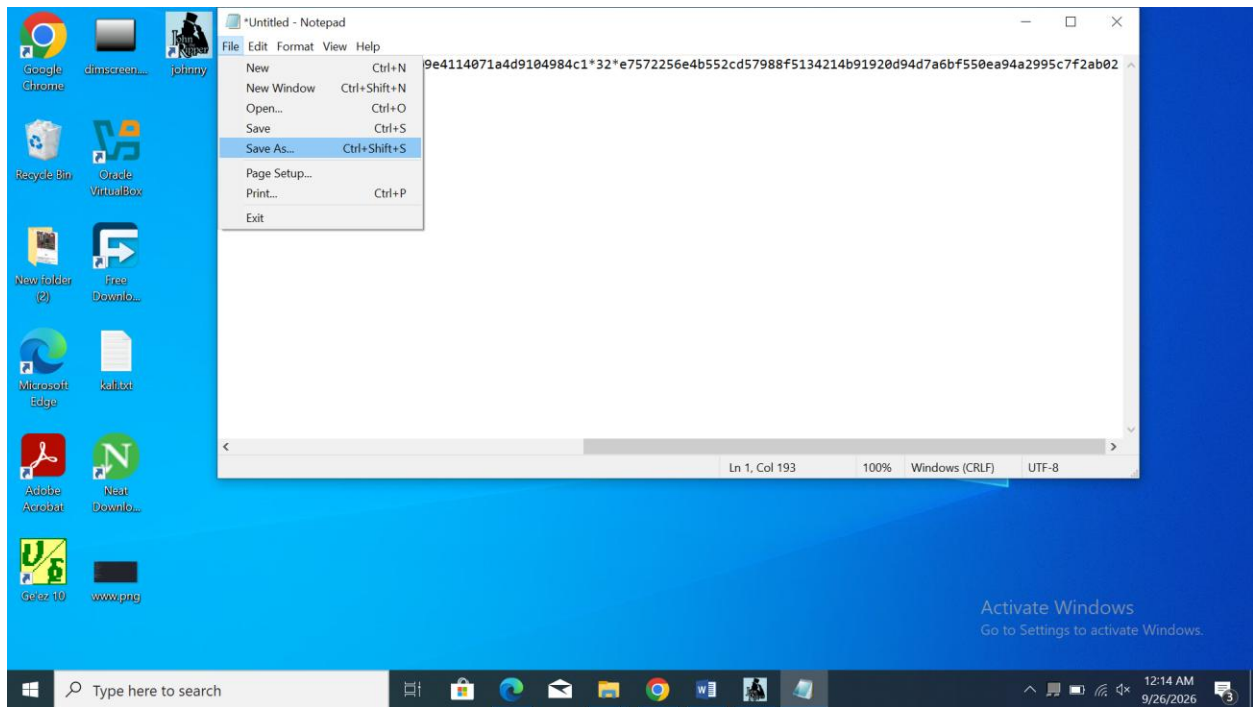
pdf2hashcat.

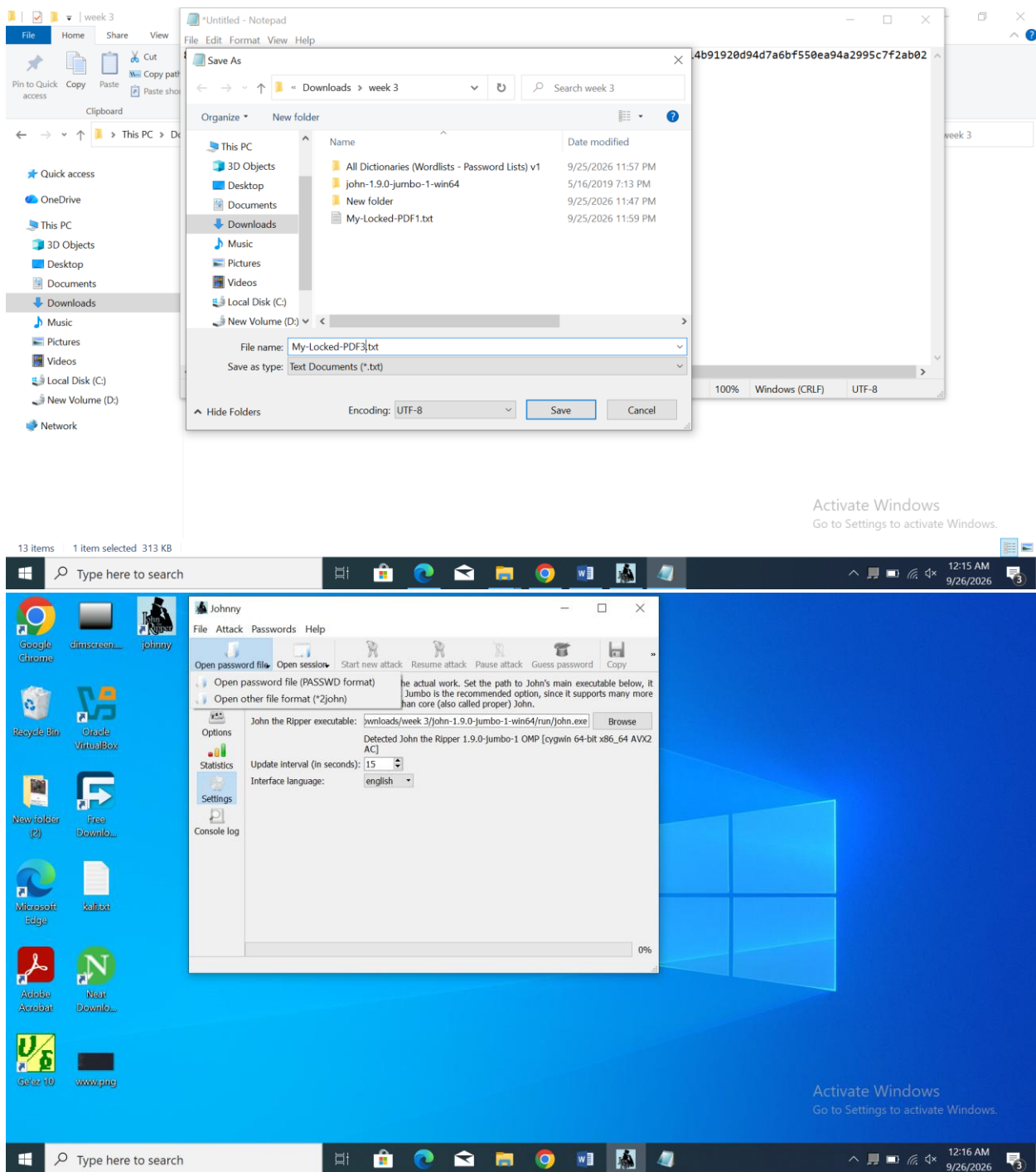
ely on our platform.

Docs, Sheets, Slides, PDF: One

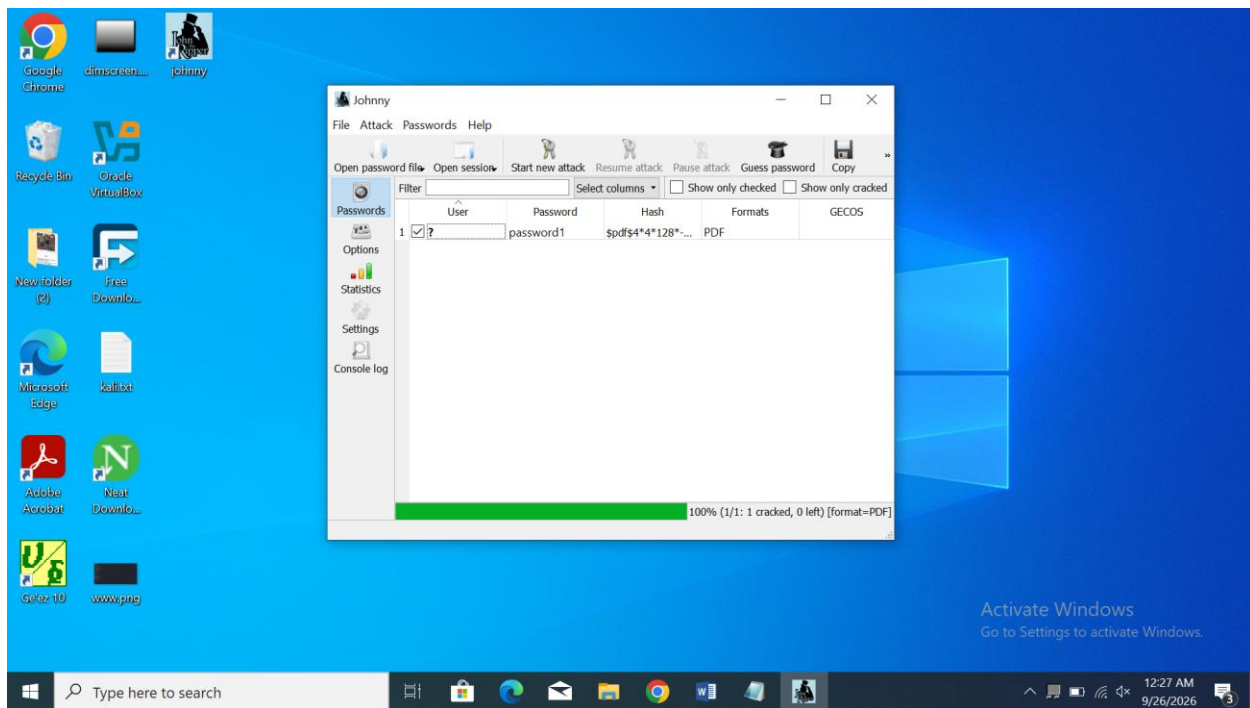
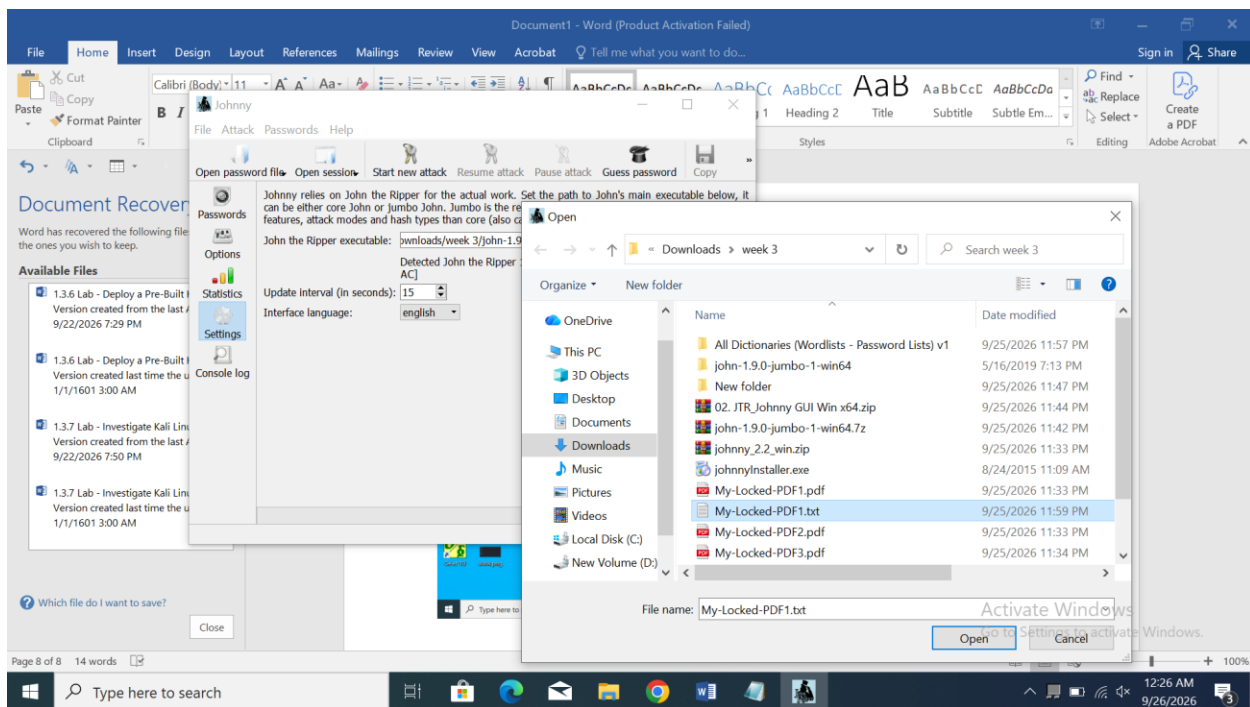
Activate Windows  
Go to Settings to activate Windows.

12:13 AM 9/26/2026









## Task 2

Crack the password of attached PDF file (My Locked PDF1.pdf) using Networkwalks tools on online.

Go to <https://networkwalks.com/hash-calculator/>

networkwalks.com/hash-calculator/



### Hash Calculator

Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF



Text

File

PDF



Click to choose a PDF, or drop one here  
Nothing is uploaded, the PDF is parsed locally

My-Locked-PDF3.pdf | 313.5 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

```
$pdf$4*4*128*-1028*1*16*34eb542eff4e1b0b32d25ce15a9a7281*32*b77872bfc9a24fb2f8  
45066283a8fc1b0021446990b9e4114071a4d9104984c1*32*e7572256e4b552cd57988f513421  
4b91920d94d7a6bf550ea94a2995c7f2ab02
```

CopyDownload

Revision: R4 Version: V4 Key Length: 128 bit

Upload a PDF to check if it is password protected and extract a crackable hash.

Free tool by Networkwalks Academy | Cybersecurity & Ethical Hacking with AI





# Hash Calculator

Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF



TextFilePDF



Click to choose a PDF, or drop one here  
Nothing is uploaded, the PDF is parsed locally

My-Locked-PDF3.pdf | 313.5 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

```
$pdf$4*4*128*-1028*1*16*...-281*32*b77872bfc9a24fb2f8  
4500...  
4b9.../2256e4b552cd57988f513421
```

CopyDownload

Revision

Ask Gemini

Search Google for "\$pdf\$4\*4\*128\*-1028\*1\*16\*..."

Print... Ctrl+P

Open in reading mode Alt+Shift+R

Translate selection to English

Use Quillbot Tools

Inspect

Upload a PDF

Crackable hash.

Security & Ethical Hacking with AI

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Go to <https://networkwalks.com/password-cracker/>

Paste the hash and click start crack

The screenshot shows the 'Password Cracker through Dictionary Attacks' interface. It includes a 'PDF HASH (\$PDF\$...)' field containing a long alphanumeric hash. Below this field are two buttons: 'Built-in list (100 passwords)' and 'Upload wordlist (.txt)'. A large pink 'START CRACKING' button is prominently displayed. The interface also features a 'Sound on' toggle and a list of instructions for using the tool.

networkwalks.com/password-cracker/

Network Walks  
DICTIONARY ATTACK LAB  
**Password Cracker through Dictionary Attacks**  
Hash every word in a wordlist and match it against a PDF password hash, the same idea John the Ripper uses.

1 Extract the \$pdf\$ hash in the Hash Calculator 2 Paste it below  
3 Run the attack, then open your PDF with the password

PDF HASH (\$PDF\$...)

\$pdf\$4\*4\*128\*-1028\*1\*16\*34eb542eff4e1b0b32d25ce15a9a7281\*32\*b77872bfc9a24fb2f845066283a8fc1b0021446990b9e4114071a4d9104984c1\*32\*e7572256e4b552cd57988f5134214b91920d94d7a6bf550ea94a2995c7f2ab02

Built-in list (100 passwords) Upload wordlist (.txt) Active: built-in list, 100 words

**START CRACKING**

Free tool by Networkwalks Academy | Cybersecurity & Ethical Hacking with AI

The screenshot shows the same interface after the cracking process is complete. A progress bar at the top indicates 'Tried: 35 / 100' and '4 pw/s' with a '35%' completion rate. A list of passwords being tested is shown, with '1qaz2wsx' marked as a match. A large green box displays the 'PASSWORD CRACKED SUCCESSFULLY' message and the password '1qaz2wsx'. Below this, a 'Copy password' button is visible. The PDF hash field remains at the bottom.

networkwalks.com/password-cracker/

Network Walks  
DICTIONARY ATTACK LAB  
**Password Cracker through Dictionary Attacks**  
Hash every word in a wordlist and match it against a PDF password hash, the same idea John the Ripper uses.

1 Extract the \$pdf\$ hash in the Hash Calculator 2 Paste it below  
3 Run the attack, then open your PDF with the password

Tried: 35 / 100 4 pw/s 35%

[+] Trying: login X  
[-] Trying: starwars X  
[-] Trying: 121212 X  
[-] Trying: bailey X  
[-] Trying: freedom X  
[-] Trying: shadow X  
[-] Trying: master X  
[-] Trying: 666666 X  
[-] Trying: superman X  
[+] MATCH 1qaz2wsx ✓

**PASSWORD CRACKED SUCCESSFULLY**

**1qaz2wsx**

This is the PDF password. Open your PDF and enter it to unlock the file.

Copy password

PDF HASH (\$PDF\$...)

\$pdf\$4\*4\*128\*-1028\*1\*16\*34eb542eff4e1b0b32d25ce15a9a7281\*32\*b77872bfc9a24fb2f845066283a8fc1b0021446990b9e4114071a4d9104984c1\*32\*e7572256e4b552cd57988f5134214b91920d94d7a6bf550ea94a2995c7f2ab02

Compiled by: Bikila Giragn

LinkedIn: <https://www.linkedin.com/in/bikila-giragn-827499241/>

